

We May Not Need Much Visual Encoding of Web Data for Question Answering

Tan Yue[†]
Wangxuan Institute of Computer
Technology, Peking University
Beijing, China
yuetan@pku.edu.cn

Qiong Wu[†]
Wangxuan Institute of Computer
Technology, Peking University
Beijing, China
wu_qiong@stu.pku.edu.cn

Dongyan Zhao^{*}
Wangxuan Institute of Computer
Technology, Peking University
National Engineering Research
Center of New Electronic Publishing
Technologies
Beijing, China
zhaody@pku.edu.cn

Abstract

With the growing scale of multimodal large language models, multimodal understanding and question answering systems have achieved remarkable progress in processing and analyzing web data. However, these improvements require excessive visual token encoding, resulting in high memory usage and computational costs. To address this limitation, we propose LightVAM, a plug-and-play lightweight visual augmented model that adaptively adjusts visual encoding according to input complexity. We construct a human-annotated multimodal complexity-aware dataset (MCA10K) covering 60 domains from web with 10,000 samples. The MCA10K is used to train a CLIP-based router that assigns instances to three processing types: simple, difficult, and OCR-augmented. For each type, LightVAM adopts distinct encoding strategies: 1) reducing image resolution for simple cases, 2) applying saliency-guided regional attention for complex visual reasoning cases, and 3) incorporating OCR-based textual enhancement for text-rich images. Experiments on multiple visual question answering benchmark datasets demonstrate that LightVAM significantly reduces visual tokens while maintaining or even improving model accuracy, providing an efficient and generalizable solution for web-scale multimodal understanding.

CCS Concepts

• **Computing methodologies** → **Computer vision; Natural language processing.**

Keywords

Multimodal Web Data, Multimodal Large Language Model, Lightweight Visual Encoding, Visual Question Answering

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[†]Equal contribution.

^{*}Corresponding author.



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1 Introduction

With the rapid advancement of multimodal large language models (MLLMs), remarkable progress has been achieved in representing joint visual-textual information, enabling these models to effectively process multimedia web data. Models based on visual-textual joint modeling, such as BLIP-2 [6] and GPT series [4], have been widely applied to web-oriented tasks, including visual question answering (VQA) and web content understanding. Leveraging large-scale web image-text alignment data, these models demonstrate strong cross-modal understanding and generation capabilities. However, as model sizes grow, the cost of training and inference increases sharply. To maintain fine-grained visual accuracy, current models encode a large number of visual tokens, which results in high GPU memory usage, significant computational overhead, and increased inference latency.

To address these challenges, researchers have explored various lightweight visual encoding methods. Early works mainly focused on improving Vision Transformer (ViT) architectures, such as DynamicViT [9], which reduces redundant visual features by dynamically pruning or selecting inter-layer tokens. Then, Token Merging (ToMe) [2] and BLIP-2's Q-Former [6] achieved end-to-end token reduction through token aggregation and visual compression, effectively alleviating redundancy during multimodal inference. More recent studies like VisionZip [14] and ATP-LLaVA [15] introduced learnable token pruning mechanisms that dynamically preserved critical information from images, significantly reducing visual token counts with minimal performance loss. Despite these advances, several key limitations remain: 1) Most general MLLMs adopt a fixed number of visual tokens, which can only be manually adjusted through hyperparameters and cannot adapt dynamically to input complexity or task requirements. 2) Although ViT-based encoders improve visual attention, they often overlook small and fine-grained visual cues, lacking a mechanism for “zoom-in and attention enhancement.” 3) Existing methods are often model-specific, showing limited transferability and generalization across different MLLMs or web-based vision tasks, which restricts their practical deployment and scalability.

Accordingly, we propose LightVAM, a plug-and-play **lightweight visual augmented model** designed to enable adaptive adjustment

and enhancement of visual information. LightVAM can be integrated into mainstream MLLMs and dynamically optimizes visual encoding according to the features of the input data. Specifically, we construct a multimodal complexity-aware dataset (MCA10K) containing over 10,000 manually annotated samples, covering multimodal web data with varying levels of complexity. Using this dataset, we train a CLIP–ALBERT fusion model (router) to assign multimodal inputs to three processing types: simple, difficult, and OCR-augmented. 1) Simple: Image resolution is reduced to minimize visual tokens; 2) Difficult: LightVAM first performs global downsampling to reduce visual tokens of the raw image, then applies saliency-guided regional attention to highlight key visual cues and integrates the enhanced regions with the raw image. 3) OCR-Augmented: Text extracted by an OCR model [1] is integrated with the reducing raw image, improving text comprehension.

Experimental results show that LightVAM consistently reduces visual tokens by more than 50% across multiple benchmark datasets and MLLMs, with the maximum reduction reaching 65.79%. Also, it maintains or even improves accuracy, achieving up to 2.27%. Without training or modifying the structure of MLLMs, LightVAM achieves generalizable and adaptive lightweight visual encoding with only 1B model size, providing an efficient and flexible solution for large-scale web-based multimodal understanding tasks.

Our main contributions are as follows: 1) We propose LightVAM, a plug-and-play lightweight visual augmented model that enables flexible integration across diverse MLLMs, demonstrating strong adaptability and generalization capabilities. 2) We construct a multimodal complexity-aware dataset (MCA10K) encompassing 10,000 samples across 60 web domains. All samples are manually annotated and cross-proofread to ensure the high quality. 3) Experimental results across multiple VQA benchmark datasets demonstrate that LightVAM can significantly reduce visual tokens while maintaining or even improving model performance, validating the efficiency and generalizability of the proposed method.

2 Methodology

To achieve lightweight and adaptive visual encoding, we propose a complexity-aware lightweight visual augmented model (LightVAM). LightVAM first assigns each input image–text pair to a specific complexity level and then processes the visual input differently. As shown in Figure 1, it consists of two main modules: 1) A multimodal complexity-aware router built on ALBERT [5] and CLIP [8] to classify sample complexity (*Simple*, *Difficult*, and *OCR-Augmented*). 2) A routing and processing module that applies distinct visual encoding strategies based on the complexity. This design enables the model to dynamically balance computational cost and accuracy during inference, achieving efficient multimodal understanding and reasoning at web scale.

2.1 Complexity-Aware Router

Given a sample (I, T, y) , where I denotes the image, T denotes the text query, and $y \in \{\text{Simple}, \text{Difficult}, \text{OCR}\}$ denotes the label, text and image encoders produce embeddings: $\mathbf{t} = \text{ALBERT}(T)$, $\mathbf{v} = \text{CLIP}(I)$, which are ℓ_2 -normalized to remove scale variance:

$$\tilde{\mathbf{t}} = \frac{\mathbf{t}}{\|\mathbf{t}\|_2 + \varepsilon}, \quad \tilde{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2 + \varepsilon}. \quad (1)$$

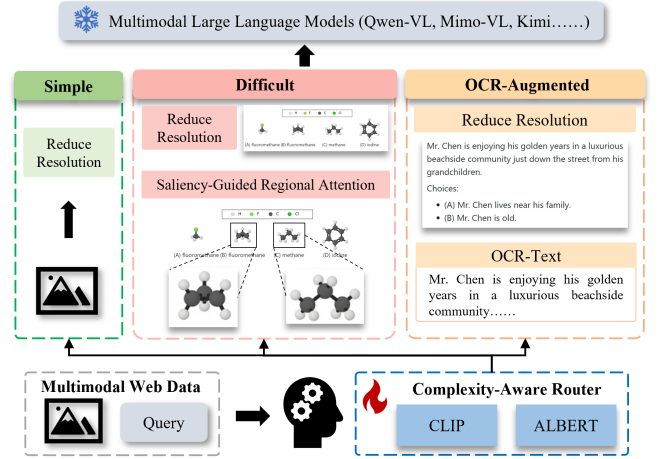


Figure 1: The proposed LightVAM model.

To characterize the text density in the image, we first construct a binary map B by applying Sobel gradients, Otsu thresholding, and morphological closing to the grayscale input. Let $B \in \{0, 255\}^{H \times W}$, where H and W denote image height and width. To estimate the number of text rows, we compute the horizontal projection $P(u) = \sum_{v=1}^W B(u, v)$ and count projection responses above an adaptive threshold τ_{proj} . The ratio of text-like pixels r and the normalized number of text lines l are defined:

$$r = \mathbb{E}_{(u,v) \in \Omega} \left[\frac{B(u, v)}{255} \right], \quad l = \frac{1}{6} \log \left(1 + \sum_{u=1}^H \mathbb{I}(P(u) > \tau_{\text{proj}}) \right). \quad (2)$$

The normalized visual, textual, and density features are concatenated: $\mathbf{z} = [\tilde{\mathbf{t}}; \tilde{\mathbf{v}}; (r, l)]$, and passed through a two-layer feed-forward network to output the probability distribution:

$$p(y|I, T) = \text{Softmax}(W_2 \text{ReLU}(W_1 \mathbf{z} + b_1) + b_2). \quad (3)$$

The model is optimized with the cross-entropy loss:

$$\mathcal{L} = - \sum_c y_c \log p(y = c|I, T). \quad (4)$$

2.2 Routing and Processing

At inference time, the trained router f_{θ^*} predicts the complexity probabilities $\mathbf{p} = f_{\theta^*}(I, T)$ and determines the processing type: $c^* = \arg \max_c p_c$. Three routing branches are designed as follows.

1) **Simple**: Reduce the resolution of images to $K \times K$.

2) **OCR-Augmented**: When dense textual regions are detected, the OCR model extracts the textual content S and concatenates it with the original question to form $T' = [T; S]$, which enhances the model's understanding of text-rich images. Although text tokens are added, the resolution of the image input is also reduced to $K \times K$.

3) **Difficult**: For visually complex samples, both the original image and automatically generated regional-attention augmented images are used as inputs. The resolution of the original image is first reduced to $K \times K$, preserving global features. Then, a text-guided relevance heatmap is computed:

$$H(x, y) = \langle \hat{\mathbf{v}}_x, y, \hat{\mathbf{t}} \rangle, \quad (5)$$

where $\hat{v}_{x,y}$ denotes the normalized visual embedding at position (x, y) and $\hat{t}$ denotes the textual embedding. An integral image enables efficient sliding-window scoring:

$$\text{score}(R) = \frac{1}{|R|} \sum_{(x,y) \in R} H(x, y), \quad (6)$$

followed by non-maximum suppression (NMS) to select the top- j salient regions $\{R_i^*\}_{i=1}^j$. Each region is cropped and resized, and the patches are concatenated:

$$I_{\text{reg}} = \mathcal{M}(\mathcal{W}(I|R_1^*), \dots, \mathcal{W}(I|R_j^*)), \quad (7)$$

where $\mathcal{W}$ denotes cropping and rescaling, and $\mathcal{M}$ indicates concatenation. Both the original (I_{ori}) and regional-attention augmented images ($I_{\text{reg}}, K \times K$) are fed into the downstream MLLMs:

$$\mathcal{X}_{\text{dif}} = \{(I_{\text{ori}}, T), (I_{\text{reg}}, T)\}. \quad (8)$$

This dual-image input maintains a low pixel budget while improving the model’s focus on informative regions. The proposed complexity-aware method integrates routing and adaptive visual processing into a unified pipeline. By dynamically selecting the three branches, the model preserves essential semantics with reduced redundant encoding, thus achieving efficient and accurate multimodal reasoning under limited computational budgets.

3 Dataset

Collection: We construct a multimodal complexity-aware dataset (MCA10K) containing 10,000 manually annotated samples to support the training of LightVAM. Samples are collected from publicly available multimodal question answering data from the web, covering 11 main domains and 60 sub-domains. The detail of the domain distribution is shown in Figure 2. During sampling, we follow three key criteria: semantic diversity, domain coverage, and task balance. (1) We prioritize samples with rich question semantics and diverse visual contexts to include various image types and reasoning requirements. (2) Within each domain, we maintain a balanced proportion of descriptive, selective, and reasoning-type questions. (3) Blurry, low-resolution, or noisy samples are filtered out to ensure visual clarity and semantic consistency. As a result, the dataset achieves broad coverage across tasks and domains, providing a solid foundation for enhancing the generalization of trained models.

Annotation: The annotation process consisted of three stages. **Stage 1: Definition of annotation criteria.** Each sample is categorized into one of three types. 1) Simple: The image contains straightforward content directly aligned with the text, and the answer can be derived through global visual understanding. 2) Difficult: The question involves complex reasoning, multi-object recognition, or cross-region information integration, requiring stronger visual reasoning. 3) OCR-Augmented: The image contains substantial textual or tabular information, and the answer depends on the semantic understanding of the text in the image. **Stage 2: Initial annotation.** Each sample is independently labeled by two annotators according to the defined criteria to minimize subjective bias. **Stage 3: Consistency verification.** Samples with inconsistent annotations are reviewed and corrected by a senior annotator. In cases of disagreement, three annotators discuss and finalize the label through a majority voting process. To evaluate the quality of annotation,



Table 1: Performance comparison on benchmark datasets.

Model	MathVista		M3CoT		SciChart	
	Acc	Tokens	Acc	Tokens	Acc	Tokens
Qwen2.5-3B	62.1	543.2	48.06	535.95	29.5	756
Δ	$\downarrow 4.3$	$\downarrow 65.79\%$	$\uparrow 0.78$	$\downarrow 54.51\%$	$\uparrow 1.15$	$\downarrow 63.27\%$
+LightVAM	57.8	185.85	48.84	243.79	30.65	277.65
Qwen2.5-7B	67.6	543.2	57.89	535.95	38.7	756
Δ	$\downarrow 0.1$	$\downarrow 65.79\%$	$\uparrow 0.61$	$\downarrow 54.51\%$	$\uparrow 2.27$	$\downarrow 63.27\%$
+LightVAM	67.5	185.85	58.5	243.79	40.97	277.65

Table 2: Ablation study results. CA=Complexity-Aware.

Model	MathVista		M3CoT	
	Acc	Tokens	Acc	Tokens
Qwen-7B w/o CA	67.3	565.78	58.01	557.16
Δ	$\downarrow 0.2$	$\uparrow 204.43\%$	$\downarrow 0.49$	$\uparrow 128.54\%$
Qwen-7B w/o OCR	66.3	168.56	57.1	227.31
Δ	$\downarrow 1.2$	$\downarrow 9.30\%$	$\downarrow 1.4$	$\downarrow 6.76\%$
Qwen-7B+LightVAM	67.5	185.85	58.5	243.79

Qwen2.5-VL-7B model, as shown in Table 2. The analysis focuses on two key components: 1) The complexity-aware (CA) routing module, which distinguishes between simple and difficult samples for parallel processing, and 2) the OCR Text Enhancement module, which improves model understanding in text-rich visual scenarios. Results show that removing CA module (w/o CA) decreases accuracy by 0.2% and 0.49% on MathVista and M3CoT, respectively, while greatly increasing the number of visual tokens by 204.43% and 128.54%. This highlights the crucial role of the CA module in reducing visual redundancy. Removing OCR enhancement (w/o OCR) leads to accuracy drops of 1.2%-1.4% on the two datasets, showing that text-aware augmentation remains essential for text-rich images. In contrast, the full LightVAM achieves the best trade-off between accuracy and token efficiency, maintaining high performance while significantly reducing visual tokens.

Model-Agnostic Adaptability. To verify the model-agnostic adaptability of LightVAM, we integrate it into two more MLLMs with distinct architectures: Mimo-VL-RL-7B and Kimi-Instruct-16/A3B, and evaluate performance on the M3CoT dataset, as shown in Table 3. Results show that LightVAM achieves stable and consistent lightweight visual encoding across different MLLMs. For Mimo-VL-RL-7B, integrating LightVAM reduces the number of visual tokens by 54.51%, with only a 0.3% decrease in accuracy. For Kimi-Instruct-16/A3B, the number of visual tokens decreases by 57.57%, while performance remains almost unchanged, dropping by only 0.1%. These findings confirm that LightVAM efficiently adapts to MLLMs with different architectures and parameter scales without any training or modification.

5 Conclusion

In this paper, we present LightVAM, a plug-and-play 1B model that enables adaptive lightweight visual encoding for MLLMs. Using the

Table 3: Model Adaptability Experiment.

Model	Mimo-VL-RL-7B		Kimi-Instruct-16/A3B	
	Acc	Tokens	Acc	Tokens
Raw	65.1	535.95	58.7	511.09
Δ	$\downarrow 0.3$	$\downarrow 54.51\%$	$\downarrow 0.1$	$\downarrow 57.57\%$
+LightVAM	64.8	243.79	58.6	216.84

MCA10K dataset and a complexity-aware router, LightVAM applies targeted encoding strategies for different input types. Extensive experiments across multiple benchmarks and MLLMs show that LightVAM significantly reduces visual tokens while maintaining or improving model accuracy. Without training or modifying MLLMs, LightVAM provides an efficient and generalizable solution for web-oriented multimodal understanding systems.

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